
DAPHNE DAQ Integration

Interface contract, DAQ data path,
calibration scans, and mitigation evidence

Manuel Arroyave

June 2026

DUNE
DEEP UNDERGROUND
NEUTRINO EXPERIMENT

 **Fermilab**

Executive Claim

INTEGRATION STATUS

DAPHNE Ethernet data taking is established in DAQ. Selected calibration scans are moving into pds-run.

EDMS 2088726 v7.0 anchors the DAQ/PDS interface; DAQ libraries own decode, with Waffles downstream.

Interface

EDMS 2088726 v7.0 is the interface reference.

Nov 2025

DAPHNE Ethernet data taking proven in the VD Coldbox.

May 2026

Selected DAPHNE settings exercised through DAQ runs.

Firmware evidence

Peak descriptors and grouped-Hermes studies support mitigation.

Integration Evidence

Formal interface

Reference the DAQ/PDS boundary through EDMS 2088726 v7.0, for both VD and HD detectors.

DAQ data path

DAPHNE Ethernet frames enter DAQ, are decoded through DAQ libraries, and are consumed downstream.

DAQ-wrapped scans

Selected DAPHNE settings are scanned by `pds-run`: every scan point becomes a DAQ run. Naturally, future calibrations become datasets within a run.

Integration scope: interface reference, DAQ formats and decode, controller/monitoring path, and run-controlled DAPHNE scans. Resource and dead-time mitigation are summarized on dedicated evidence slides.

Interface Contract

FORMAL REFERENCE

**DUNE Interface Control Document: DAQ
& Photon Detector System (PDS)**

EDMS 2088726 v7.0

<https://edms.cern.ch/document/2088726/7>

This is the common DAQ/PDS boundary for the readout, timing, control, monitoring, and data-format discussion.

Interface scope

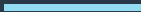
- ▶ DAQ/PDS boundary and responsibilities.
- ▶ DAPHNE-to-DAQ 10 GbE UDP data interface.
- ▶ DAQ header and DAPHNE header responsibilities.
- ▶ Timing provenance and calibration-command tagging.
- ▶ CCM configuration, control, monitoring, and error reporting path.



SECTION

DAQ Baseline

The implementation maps the ICD data/control boundary into DAQ packages, decode utilities, and monitoring



Baseline Data Path



Format

C++ classes describe the wire layout and enforce bit-field boundaries.

Decode

Python bindings expose arrays for analysis without rewriting the unpacker.

Waffles

Waffles consumes the decoded arrays for calibration and validation workflows.

Changed DAQ Repositories

fdetdataformats

Frame definitions and unit tests for DAPHNE Ethernet data.

Peak descriptor support included.

rawdatautils

DAQ-side unpackers expose ADC, timestamp, channel, stream, and peak arrays. Used by validation and Waffles.

daphnemodules

DAPHNE V3 controller path for configuration and operational monitoring.

Includes V3 GeneralInfo opmon.

Together: format, decode, and controller changes are separate DAQ packages, but they form one DAPHNE readout chain.

Data-Format Contract

DAPHNEethFrame

Single-channel frame: Ethernet header, DAPHNE metadata, five peak descriptors, and 1024 packed 14-bit ADC samples.

Peak descriptors

New firmware information is now persisted in the frame and exposed through DAQ formats and Python decode utilities.

DAPHNEethStreamFrame

Four-channel stream frame with 280 samples per channel packed as 14-bit ADCs.

Contract checks

- ▶ ICD-backed constants: ADC width, sample count, channel count.
- ▶ Accessors validate channel, sample, and peak indices.
- ▶ Unit tests round-trip ADC and peak fields.
- ▶ Bindings register triggered and stream types.

Decode Contract

RAW DATA UTILS

`rawdatautils` is not a second format definition. It reuses `fdetdataformats` and exposes arrays consumed by Waffles and validation scripts:

- ▶ ADC, timestamp, and channel arrays
- ▶ stream-frame equivalents
- ▶ peak descriptor arrays for triggered frames



Controller And Monitoring Contract

daphnemoDules V3 controller

Configures the DAPHNE V3 analog chain, reads counters, and publishes opmon in DAQ.

Important release finding

fddaq-v5.5.0-a9-1 ships 3.0.1. V3 GeneralInfo opmon appears in 3.0.2 on patch/fddaq-v5.5.x.

ReadTriggerCounters



per-channel
threshold, record, busy, full



ReadGeneralInfo
bias, power, temp



Kafka opmon
shifter view

Dead-Time Mitigation Evidence

OPERATING POINT

4.7 kHz/channel

rounded comparison

Main

7%

1024-tick waveform

Mitigation

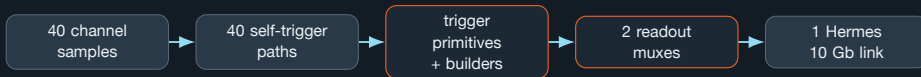
2.5%

512-tick waveform

Main caveat

The lower point assumes waveform records are restricted to 512 ticks.

Main HDL topology

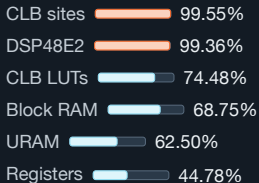


Mitigation HDL topology



K26 Resource Mitigation Evidence

BASELINE, OUTPUT-2C337B5



GROUPED CANDIDATE, E5E2D70



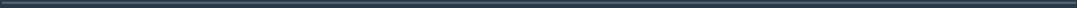
The grouped candidate keeps LUT, BRAM, and URAM below device limits and removes the baseline DSP pressure. The remaining qualification is readout behavior, not FPGA fit or route legality.



SECTION

Calibration Path Into DAQ

Selected scans already generate DAQ configurations, DAQ runs, and reviewable datasets



Calibration Scans As DAQ Artifacts

CURRENT STATUS

Selected calibration variables now generate DAQ configurations and run-associated datasets.

Calibration probes

Probing identified variables, ranges, and target settings worth moving into DAQ.

DAQ wrapper

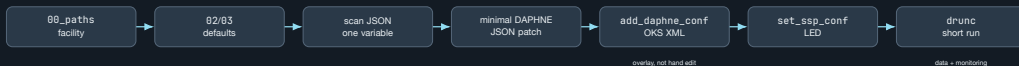
pds-run drafts the DAQ workflow: generate the overlay and delegate each scan point to drunc.

Dataset path

The result is DAQ data and monitoring tied to the configuration that produced it.

Integration point: the mechanics are functional now; the endpoint is the same scan semantics implemented natively inside DAQ.

Draft DAQ Workflow For Variable Scans



Scan plan

One variable, explicit values, and optional SSP LED settings.

DAQ config

Minimal DAPHNE overlay generated for the scan point.

drunc wrapper

Each scan point delegates to DAQ run control.

Dataset

Data and monitoring are tied to the DAQ run.

Functional Scan Coverage

Self-trigger

Threshold values become short DAQ runs with changed trigger settings.

AFE response

Attenuation, bias, offset, and trim become generated overlays plus run data.

LED / SSP

LED intensity and mask are coordinated with DAQ data taking.

Plan mode

Generated XML and run-control actions can be inspected before execution.

The important result is not command syntax: variable scans already exercise the DAQ configuration path and produce run-associated datasets with the current system.

Operational Environment

Login path

marroyav@np04-srv-017 then
np04daq@np04-srv-024; use tmux session
np02-vdcb.

Workarea

/nfs/sw/dunedaq/
dunedaq-fddaq-v5.5.0-dev-pds.
Source setup_for_run.sh.

Proxy state

Use source ~/bin/web_proxy.sh only for
package downloads. Disable for running with -u.

Hardware target

Board identity and IP must be checked in the active
JSON/XML; target DAPHNE selection is a
configuration requirement, not an assumption.



SECTION

Validation And Review

The success condition is both software cleanliness and live DAQ behavior



Live Validation Result

Run 44355

Default release without the daphnemoDules override: data taking worked, but V3 monitoring was not visible.

Run 44356

Clean daphnemoDules 3.0.2 override: two-minute TEST run succeeded at 20 Hz.

Firmware

Peak descriptors were added in firmware and then carried into the DAQ data-format contract.

Conclusion

Monitoring loss was traced to the release package version, not to run control.

Final tested state: sourcecode/daphnemoDules commit 13289b5 on patch/fddaq-v5.5.x; DAPHNE_V3_ROUTE unset; monitoring visible.

Live pds-run Scan Exercise

SCAN MODES EXERCISED

Self-trigger	thresholds: 11, 10
AFE attenuation	all attenuators: 2200, 2300
Offset	selected channels: 2260, 2275
Trim	selected channels: 1, 0
AFE bias	AFE0: 10, 0; others fixed at 0
LED / SSP	intensity: 3000, 4095; mask 4

Run span: 44357–44368; each point generated a DAQ configuration and delegated execution to `drunc`.

RESULT

Successful scan commands exited with status 0.

`xmlLint -noout` passed after each XML update.

No successful-run traceback or `drunc` command failure was found.

Evidence location

`np04-srv-024`; branch `vdcb-daq-scans`.
Logs: `/tmp/pds_vdcb_live_*`.

Live Exercise Findings

Finding	Consequence
SSP object name	Active XML uses np02-ssp; scan configs must not silently rely on the old np02-ssp-on default.
Execution seed	99_apply_fe_min_v2_subset.json is plan-safe but not execution-safe; add_daphne_conf requires detector_id.
Zero/default XML	trim=0, v_bias=0, and bias_ctrl=0 can be omitted by add_daphne_conf; absence of these attributes is not by itself a run failure.
Post-scan state	The exercise left SSP at channel_mask=4, pulse1_width=3; scan plans should state whether to restore the baseline.

Conclusion: the DAQ-wrapped scan loop is operational for the tested modes; remaining cleanup is configuration hygiene, explicit target naming, and post-scan restoration policy.

Summary

ESTABLISHED FOR PRR

Interface: EDMS 2088726 v7.0 anchors the DAQ/PDS boundary for VD and HD.

DAQ path: DAPHNE Ethernet data taking, format ownership, and decode utilities have been exercised.

Calibration path: selected DAPHNE settings can be scanned through `pds-run`; each point becomes a DAQ run.

BOUNDED QUALIFICATION

Dead time: mitigation evidence gives 7% to 2.5% at 4.7 kHz/channel, with 512-tick records.

Resources: grouped Hermes strategy addresses the K26 fit pressure.

Next check: Hermes readout, backpressure, and real-trigger stress.